SOC Detection Engineering, Alert Triage & SOC Operations

Week 1: SIEM Architecture & Log Onboarding

Lab report | Elastic Stack (Fleet + Kibana) | October 2026

1. Summary

This week I built the SIEM foundation for the lab on the Elastic Stack. A single Ubuntu server (ubuntu-siem) runs Fleet Server, Elasticsearch and Kibana. Three endpoints (Windows 10, Windows Server 2019 and Kali Linux) are enrolled as Fleet-managed Elastic Agents, and a FortiGate VM sends its logs to a file on the Ubuntu server that Filebeat reads.

All four agents show Healthy in Fleet (version 8.19.23) and Kibana Discover returns documents for each source. The report covers the five Week 1 deliverables: the architecture diagram, the data source inventory, the log onboarding evidence, the normalized field mapping and the retention plan.

What is verified. Windows Security logons (event.code 4624, dataset system.security) on the Windows Server, Sysmon process creation (event.code 1, dataset windows.sysmon_operational) on the Windows 10 machine, agent logs from Kali, and the FortiGate log file. Still to verify: Kali auditd events and the parsed FortiGate fields (section 9).

Changes during the week. The Windows 10 machine was renamed from DESKTOP-DDM5DBD to IT-machine (same agent ID). The Windows Server address moved from 192.168.1.91 to 192.168.1.116 (DHCP). The retention plan is based on measured disk growth: about 4.2 MB per minute, which is about 5.9 GB per day.

2. Week 1 Requirements and Where to Find Them

Required deliverable	What I produced	Section
SIEM architecture diagram	Architecture diagram of the lab with a medium-organization design	Section 4 (Figure 1)
Data source inventory	Inventory of endpoint, network, identity, cloud and application sources	Section 5
Log onboarding evidence	Fleet, agent enrollment and Discover screenshots for each source	Section 6
Normalized field mapping	Raw field to ECS field tables for Windows, Linux and FortiGate	Section 7
Retention plan	Measured disk growth, capacity table, lab ILM policy, target design	Section 8

3. Lab Environment

Everything runs as virtual machines in VMware Workstation on one host, on the 192.168.1.0/24 network. The values below were read from the screenshots in this report.

Machine	Hostname	IP address	Fleet policy	Role
Ubuntu server	ubuntu-siem	192.168.1.84	Fleet Server Policy	Fleet Server (8220), Elasticsearch, Kibana (5601), Filebeat for the FortiGate file
Windows 10	IT-machine (was DESKTOP-DDM5DBD)	not captured	Windows10_VM	Endpoint, workstation telemetry, Sysmon
Windows Server 2019	WIN-MKSP5LUC3TN	192.168.1.116 (was .91 on 22 Oct)	WindowsServer_VM	Endpoint, server telemetry
Kali Linux	kali	192.168.1.96	KaliLinux_VM	Endpoint, Linux telemetry, attacker box for later weeks
FortiGate VM	FGT-VM64	192.168.1.167	not an agent	Network firewall, source of network logs

The Windows Server changed address between 22 and 30 October (192.168.1.91 to 192.168.1.116), so the lab is on DHCP. Static addresses are needed so the agents do not lose the Fleet Server.

4. SIEM Architecture

The design target is a medium organization (a few hundred endpoints, a handful of servers, one perimeter firewall). The lab is a small copy of that design with every layer present once.

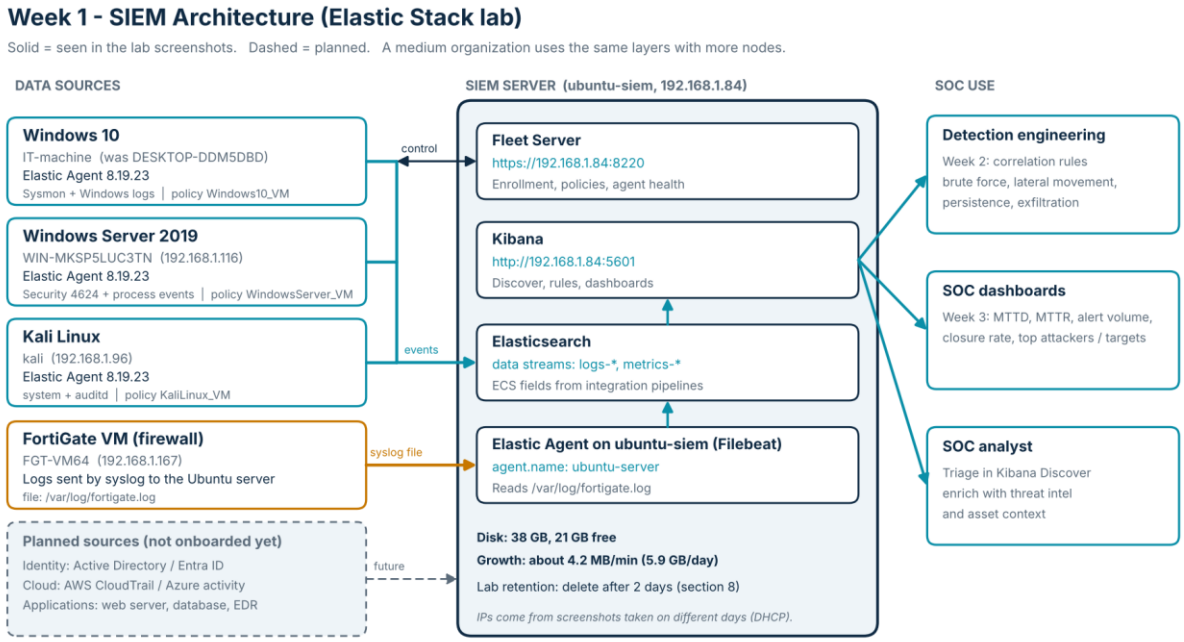


Figure 1. SIEM architecture for the lab. Solid arrows are paths seen in the screenshots; the dashed box shows sources planned for the full design.

4.1 How the data flows

- **Enrollment and policy.** Each agent enrolls to Fleet Server at https://192.168.1.84:8220. The Windows Server screenshot shows Test-NetConnection to port 8220 succeeding before enrollment.
- **Endpoint events.** The agent ships logs from the policy's integrations to Elasticsearch, where they land in data streams (for example logs-system.security-default and logs-windows.sysmon_operational-default) with ECS fields.

- **Network events.** FortiGate writes its logs to /var/log/fortigate.log on the Ubuntu server. The agent on that server reads the file with a Filebeat input (log.file.path and agent.type: filebeat are visible in Discover).
- **Analysis.** Analysts work in Kibana: Discover for hunting and triage, then rules and dashboards.

4.2 Design choices for a medium organization

Layer	Lab	Production design
Management	Fleet Server on the Ubuntu VM	2 Fleet Servers behind a load balancer, separate from Elasticsearch
Storage / search	Single Elasticsearch node, 38 GB disk	3 nodes minimum: hot/warm tiers, 1 replica, snapshots to object storage
Collection	Elastic Agent managed by Fleet; Filebeat input for the FortiGate file	Agents everywhere, syslog collector for network devices, policy per OS role
Security	Enrollment with --insecure (self-signed certificate)	Certificates from an internal CA, API keys per policy, role-based access in Kibana
Addressing	DHCP (addresses changed)	Static or reserved addresses for the SIEM and all agents

5. Data Source Inventory

The assignment asks for endpoints, network, identity, cloud and applications. Status shows what is onboarded in the lab and what is only part of the design.

Category	Source	Key log types	Collection method	Status
Endpoint	Windows 10 (IT-machine)	Sysmon operational (process creation), Windows event logs	Elastic Agent: system-2, windows-10	Onboarded, Sysmon verified
Endpoint	Windows Server 2019 (WIN-MKSP5LUC3TN)	Windows Security (logons 4624), process creation, host metrics	Elastic Agent: system-3, WindowsServer-19	Onboarded, Security verified
Endpoint	Kali Linux (kali)	auth and system logs, Linux audit records, host metrics	Elastic Agent: system-4, auditd-1	Onboarded; auditd to verify
Endpoint	Ubuntu server (ubuntu-siem)	Agent and stack logs, host metrics	Elastic Agent (Fleet Server Policy)	Onboarded
Network	FortiGate VM	Traffic, UTM, event and VPN logs	Syslog to file, read by Filebeat	Onboarded; parsing to verify
Identity	Active Directory / Entra ID	Logons, group changes, Kerberos, directory changes	Domain controller agent / Entra connector	Planned
Cloud	AWS CloudTrail, Azure activity logs	API calls, console logins, config changes	Elastic cloud integrations	Planned
Application	Web server, database, EDR	Access and error logs, query audit, endpoint alerts	Agent integrations or log files	Planned

6. Log Onboarding Evidence

6.1 Fleet agents overview

Fleet lists four agents, all Healthy, all on version 8.19.23, each with a policy and a recent check-in (19 to 38 seconds ago when the screenshot was taken). The left side of the screenshot shows the VMware library with the same VMs.

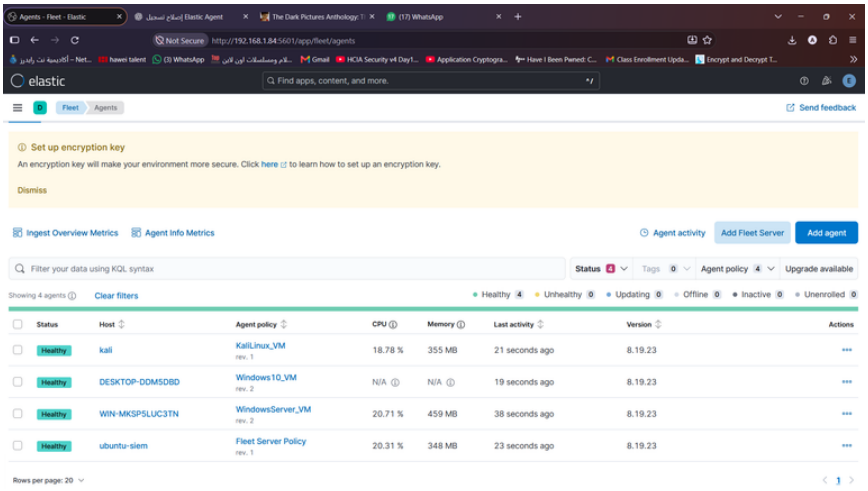


Figure 2. Fleet > Agents: kali, DESKTOP-DDM5DBD (now IT-machine), WIN-MKSP5LUC3TN and ubuntu-siem, all Healthy.

6.2 Kali Linux

The agent was installed from the elastic-agent-8.19.23-linux-x86_64 folder and enrolled with the enroll command against 192.168.1.84:8220. The terminal ends with "Successfully enrolled the Elastic Agent" and ifconfig shows eth0 at 192.168.1.96.

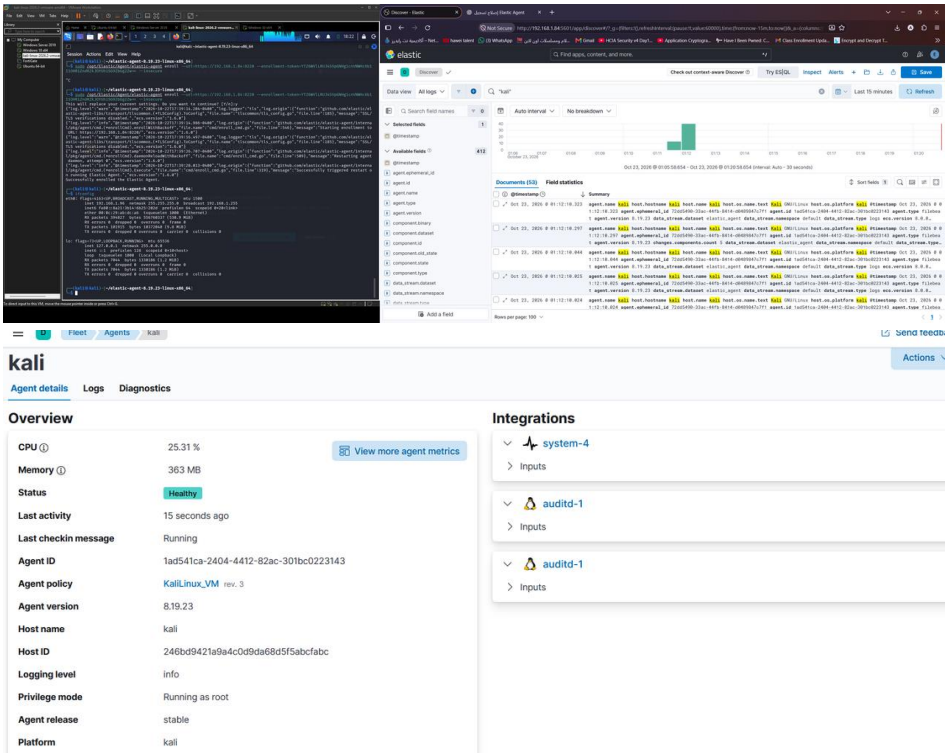


Figure 3. Top: enrollment on Kali (left) and Discover, query "kali", 53 documents (right). Bottom: Kali agent details, Healthy, policy KaliLinux_VM (rev. 3), integrations system-4 and auditd-1.

The documents carry agent.name, host.hostname, host.os.platform, agent.type (filebeat), agent.version 8.19.23, data_stream.dataset elastic_agent and ecs.version 8.0.0. These are the

agent's own logs, so the agent ships data; auditd collection is still to verify. The auditd-1 integration is listed twice on the agent page, which looks like a duplicate to clean up.

6.3 Windows 10 (IT-machine)

Enrollment was done from an elevated PowerShell with elastic-agent.exe install against https://192.168.1.84:8220 and finished with "Elastic Agent has been successfully installed". The first Discover query returned 2 process documents.

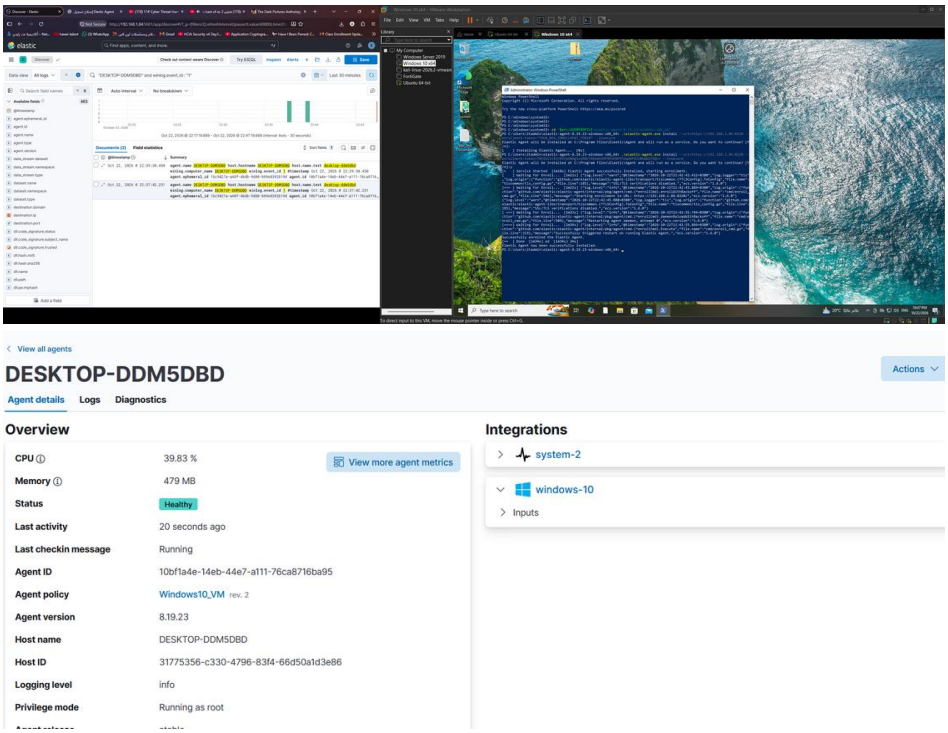


Figure 4. Top: Discover (left) and install output (right) for DESKTOP-DDM5DBD. Bottom: agent details, Healthy, policy Windows10_VM (rev. 2), integrations system-2 and windows-10.

The machine was later renamed to IT-machine. The agent ID is the same (10bf1a4e-14eb-44e7-a111-76ca8716ba95) in the Fleet detail page and in the new Discover documents, so it is the same agent under a new hostname.

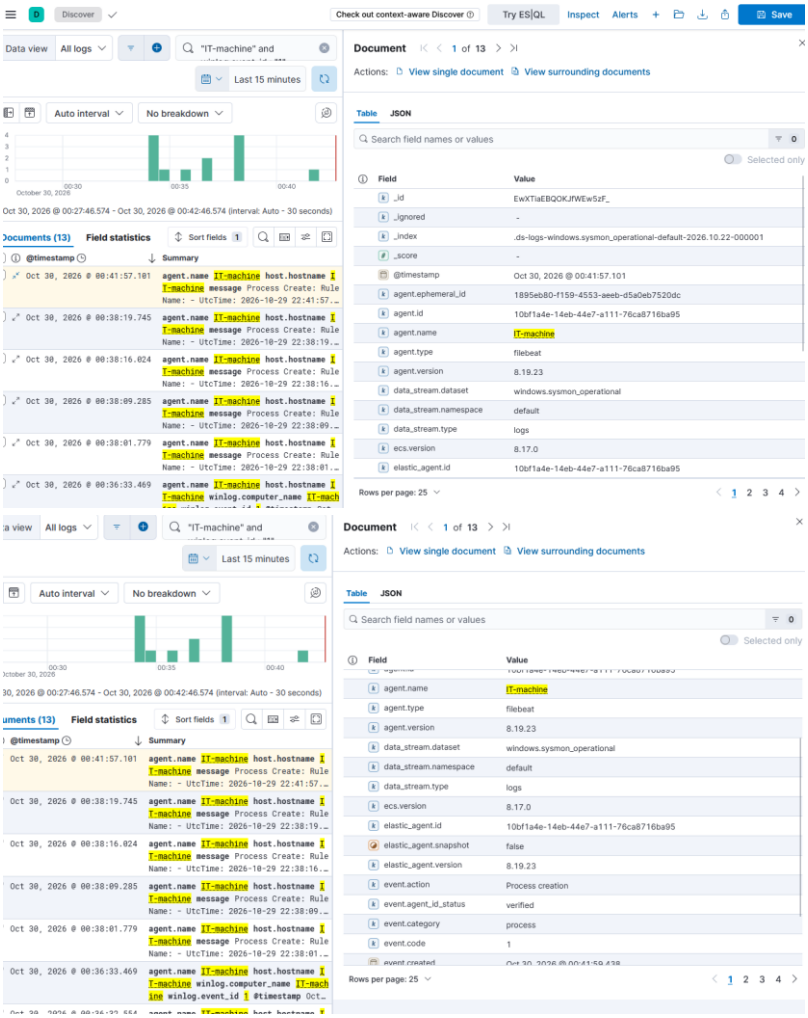


Figure 5. Sysmon process creation on IT-machine. Query "IT-machine" and winlog.event_id: "1", last 15 minutes, 13 documents. Top: index and data stream fields. Bottom: the same document scrolled down to the event fields.

Fields seen: data_stream.dataset windows.sysmon_operational, event.code 1, event.action "Process creation", event.category process, agent.type filebeat, ecs.version 8.17.0, in the index .ds-logs-windows.sysmon_operational-default-2026.10.22-000001.

6.4 Windows Server 2019

Before enrolling, Test-NetConnection 192.168.1.84 -Port 8220 returned TcpTestSucceeded: True. After enrollment, Restart-Service and the status command report both fleet and elastic-agent as HEALTHY.

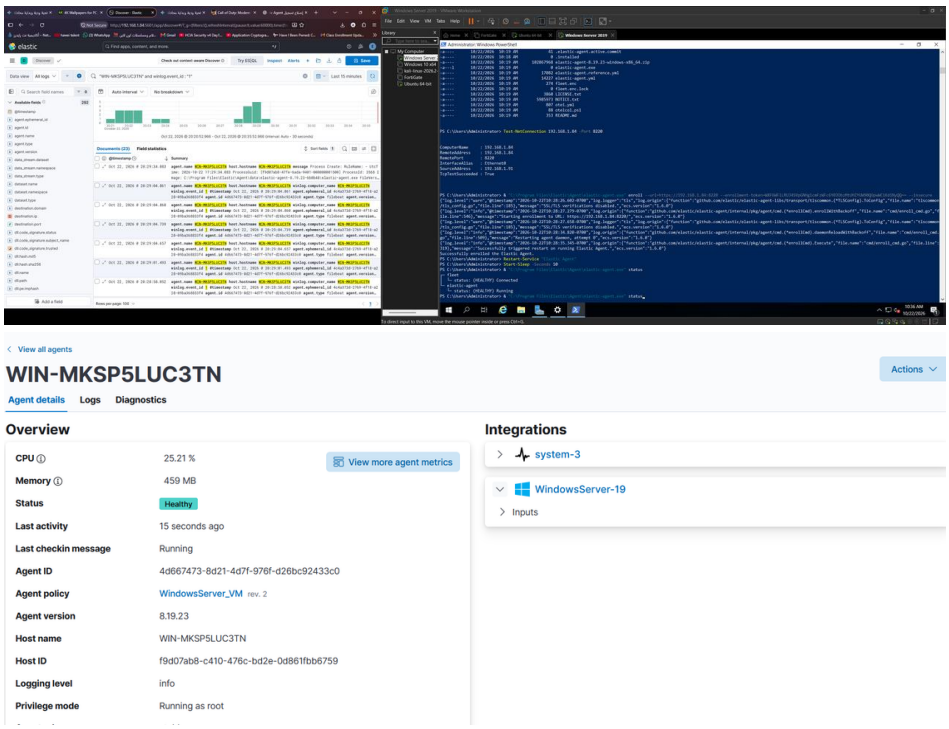


Figure 6. Top: Discover with 23 documents (left) and connectivity test, enrollment and status (right). Bottom: agent details, Healthy, policy WindowsServer_VM (rev. 2), integrations system-3 and WindowsServer-19.

Security logs are collected as well. The query on the machine name returns 70 documents in 15 minutes from the data stream logs-system.security-default.

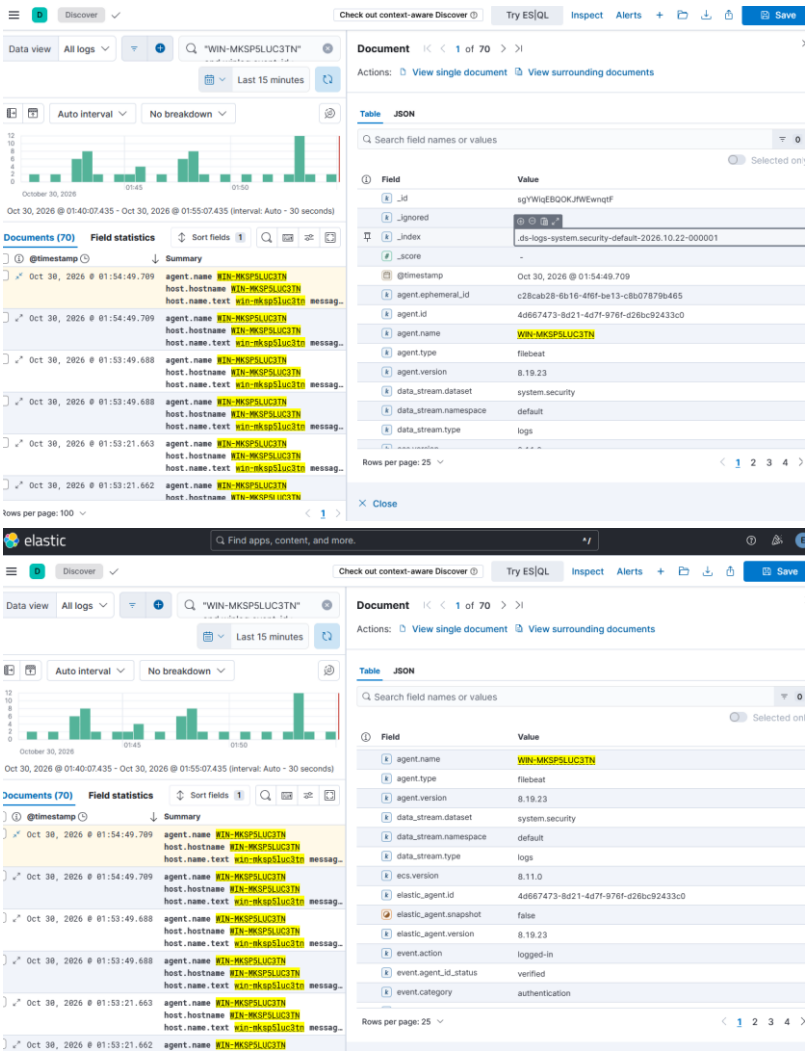


Figure 7. Windows Security events on WIN-MKSP5LUC3TN, 30 October. Top: index and data stream fields. Bottom: the same document scrolled down to the event fields.

Fields seen: data_stream.dataset system.security, event.action logged-in, event.category authentication, agent.type filebeat, ecs.version 8.11.0, in the index .ds-logs-system.security-default-2026.10.22-000001. So logon events are verified for this server.

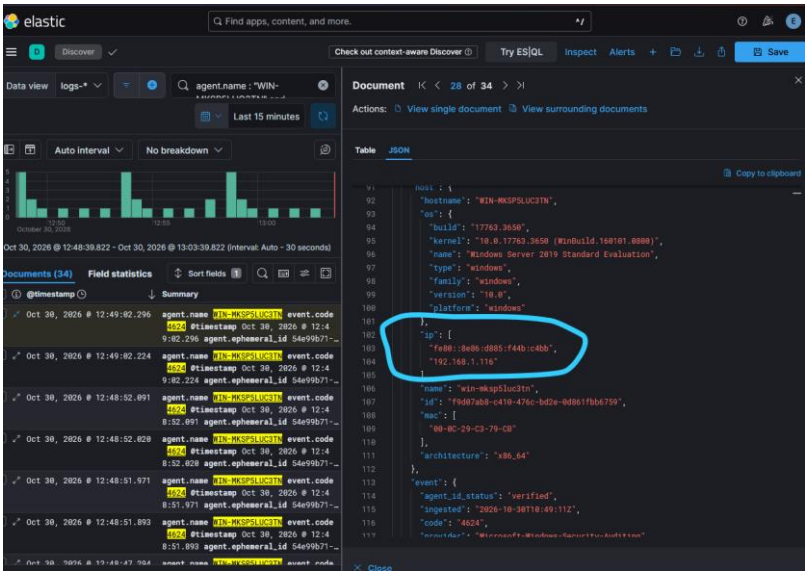


Figure 8. JSON view of a logon document (event.code 4624). The circled host.ip shows the current address 192.168.1.116. 34 documents, 12:48 to 13:03.

The same document shows host.mac 00-0C-29-C3-79-CB and host.os.name Windows Server 2019 Standard Evaluation (build 17763.3650). On 22 October this machine was at 192.168.1.91, so its address changed. The agent ID (4d667473-8d21-4d7f-976f-d26bc92433c0) and host ID are the same as before.

6.5 FortiGate

The FortiGate dashboard (FGT-VM64, 192.168.1.167) is on the right. On the left, Discover with the query "fortigate.log" returns 26 documents with log.file.path /var/log/fortigate.log, agent.name ubuntu-server and agent.type filebeat, so the logs reach Elasticsearch through the file input on the Ubuntu server.

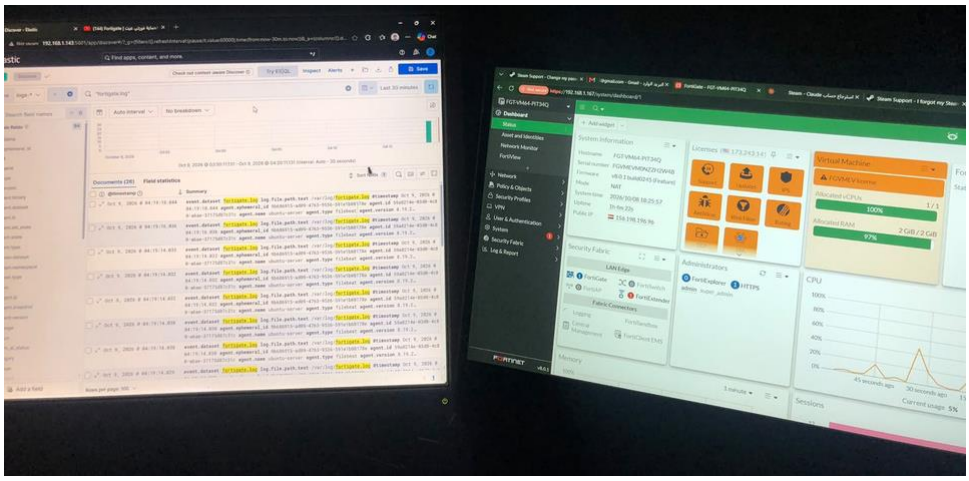


Figure 9. FortiGate dashboard (right) and Discover on fortigate.log (left, 8 October capture).

This capture is from 8 October and the summary shows file and agent fields but not the parsed firewall fields (source.ip, destination.ip, action). I need a fresh capture with those fields expanded (section 9).

6.6 Issue found and fixed: stopped agent service on Windows 10

While enrolling Windows 10 the Elastic Agent service was Stopped and Start-Service failed. The fix, visible in the screenshot, was to set the startup type to Automatic, start the service again and confirm with elastic-agent.exe status that the agent reports HEALTHY.

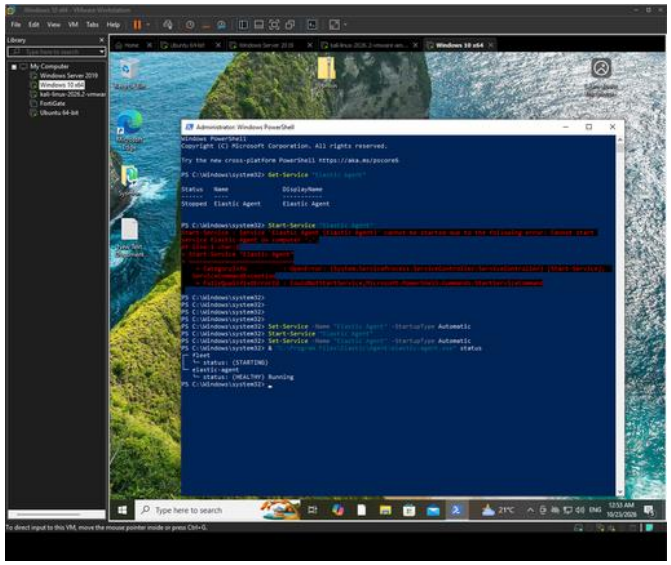


Figure 10. PowerShell on Windows 10: service Stopped, Start-Service error, Set-Service to Automatic, then status shows (HEALTHY).

7. Normalized Field Mapping

Elastic normalizes events to the Elastic Common Schema (ECS) so one query or rule works across sources. The integrations do most of this in their ingest pipelines. The tables list the mapping I will rely on for the detection rules. The Observed column says whether I saw the target field in a Discover screenshot.

7.1 Common fields (all sources)

Raw / source field	ECS field	Observed	Used for
event time	@timestamp	Yes	Timeline, ordering, rule windows
agent hostname	host.name, host.hostname	Yes	Asset identification
agent name / type / version	agent.name, agent.type, agent.version	Yes	Collector health
integration dataset	data_stream.dataset / .namespace / .type	Yes	Routing, retention, filters
event identity	event.code, event.action, event.category	Yes	Rule selection
host address / MAC	host.ip, host.mac	Yes	Asset context
OS name / platform	host.os.name.text, host.os.platform	Yes	OS-specific rules
file path read by Filebeat	log.file.path	Yes	FortiGate file source
schema version	ecs.version	Yes	Field compatibility

ecs.version is not the same everywhere, so fields must be checked per dataset:

Dataset	Data stream (index seen)	ecs.version	Source
elastic_agent	logs-elastic_agent-default	8.0.0	Kali, agent's own logs
system.security	logs-system.security-default (2026.10.22-000001)	8.11.0	Windows Server, logons
windows.sysmon_operational	logs-windows.sysmon_operational-default (2026.10.22-000001)	8.17.0	Windows 10, Sysmon

7.2 Windows event logs

Raw Windows field	ECS / normalized field	Observed	Used for
Computer / winlog.computer_name	host.name	Yes	Asset context
EventID	event.code, winlog.event_id	Yes (4624, 1)	Rule selection
logon success	event.action: logged-in, event.category: authentication	Yes	Logon rules
process creation (Sysmon 1)	event.action: Process creation, event.category: process	Yes	Process rules
TargetUserName / SubjectUserName	user.name, user.target.name	Yes	Brute force, suspicious login
IpAddress / WorkstationName	source.ip, source.domain	Yes	Logon source
LogonType	winlog.logon.type	Yes	Remote vs local logon
Image (Sysmon)	process.executable	Yes	Process rules
CommandLine	process.command_line	Yes	Suspicious command detection
ParentImage / ParentCommandLine	process.parent.executable, process.parent.command_line	Yes	Parent-child chains

Raw Windows field	ECS / normalized field	Observed	Used for
Hashes	process.hash.*, file.hash.*	Yes	Threat intel match

7.3 Linux (system and auditd)

Raw Linux field	ECS / normalized field	Observed	Used for
syslog host	host.name, host.hostname	Yes	Asset context
auth log user	user.name	Yes	SSH and sudo activity
auth log source address	source.ip	Yes	Brute force on SSH
auditd syscall / key	auditd.data.syscall, event.action	Yes	Persistence and privilege changes
auditd exe / comm	process.executable, process.name	Yes	Process context
auditd file path	file.path	Yes	Sensitive file access

7.4 FortiGate

The current input reads the file as a log file. For parsed firewall fields the data needs the Fortinet FortiGate integration (or an ingest pipeline), which maps the raw key=value fields below.

Raw FortiGate field	ECS / normalized field	Observed	Used for
srcip / srcport	source.ip, source.port	Yes	Attacker and scanner view
dstip / dstport	destination.ip, destination.port	Yes	Targeted hosts and services
proto	network.transport / network.iana_number	Yes	Protocol filter
action	event.action, event.outcome	Yes	Allowed vs blocked

8. Retention Plan

Retention depends on how fast data arrives and how much disk there is, so I measured both on ubuntu-siem instead of guessing.

8.1 Measured disk growth

I ran du -sh on /var/lib/elasticsearch twice, about 43 seconds apart, and df -h for the disk.

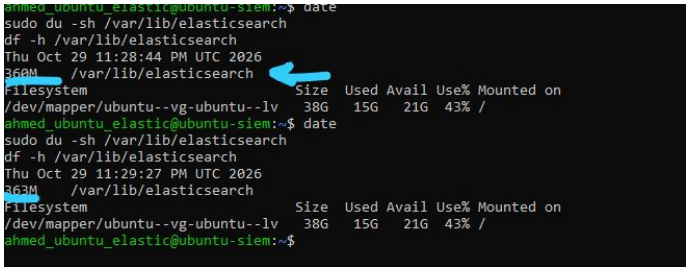


Figure 11. du -sh /var/lib/elasticsearch: 360M at 23:28:44 UTC and 363M at 23:29:27 UTC on 29 October. df -h: 38G disk, 15G used, 21G available (43%).

Item	Value
Folder measured	/var/lib/elasticsearch (Elasticsearch data)
Sample 1 / sample 2	360 MB at 23:28:44 UTC / 363 MB at 23:29:27 UTC
Interval	43 seconds
Growth	3 MB in 43 s
Per minute	3 / 43 x 60 = 4.19 MB per minute
Per hour	4.19 x 60 = 251 MB per hour
Per day	251 x 24 = 6,028 MB = 5.9 GB per day
Per source	about 1.5 GB per day each (4 sources: 3 agent hosts and the Ubuntu server with FortiGate)
Rounding range	du rounds to whole MB, so the real growth is 2 to 4 MB: 3.9 to 7.8 GB per day

Sanity check: the whole folder held only 360 MB on 29 October although the indices were created on 22 October. At 5.9 GB per day it would be about 41.2 GB after a week. So the 43-second sample was taken during a busy period and is a worst case, not the average. I plan with the worst case and will re-measure every hour for 24 hours to get the real average.

8.2 What fits on this disk

Disk needed = daily ingest x days x (1 + replicas) x 1.15 overhead. There is one node and no replicas. Elasticsearch stops allocating at 85 to 95 percent disk use (default watermarks), so usage must stay below 85 percent.

Retention	Data	With 15% overhead	Disk used after (of 38 GB)	Fits?
1 day	5.9 GB	6.8 GB	57%	Yes
2 days	11.8 GB	13.5 GB	75%	Yes
3 days	17.7 GB	20.3 GB	93%	No
7 days	41.2 GB	47.4 GB	164%	No
30 days	176.6 GB	203.1 GB	574%	No
90 days	529.8 GB	609.3 GB	1643%	No

At the worst-case rate this disk holds 2 days safely. Three days already pass the high watermark and 7, 30 or 90 days need a much bigger disk (about 47 GB for 7 days, 203 GB for 30 days, 609 GB for 90 days).

8.3 Lab retention policy

For the lab I will use one ILM policy on all logs-* data streams: roll over every day and delete after 2 days. That uses about 13.5 GB at the worst case and keeps the disk near 75 percent. If the 24-hour measurement shows a lower average, I will raise the delete age.

```
PUT _ilm/policy/soc-lab-2d
{ "policy": { "phases": {
  "hot":    { "actions": { "rollover": { "max_age": "1d" } } },
  "delete": { "min_age": "2d", "actions": { "delete": {} } }
} } }
```

To reduce volume I can also drop the elastic_agent logs, filter noisy Sysmon events and raise the metrics collection interval.

8.4 Target design for a medium organization

With enough disk the same plan becomes tiered: hot 0 to 7 days, warm 8 to 30 days, cold 31 to 90 days (searchable snapshot), delete after 90 days. Per source:

Source	Lab setting	Target for a medium organization
Windows Security / Sysmon	2 days	90 days searchable, 1 year archive
Linux auth / auditd	2 days	90 days searchable, 1 year archive
FortiGate traffic logs	2 days	30 days searchable, 180 days archive (high volume)
FortiGate security / UTM events	2 days	90 days searchable, 1 year archive
Agent / stack logs	2 days	14 days, no archive

9. Validation Status and Open Items

Check	Status	Evidence / note
All four agents enrolled and Healthy	Done	Fleet Agents overview, agent detail pages
Fleet Server reachable on 8220	Done	Test-NetConnection on Windows Server, enrollment output
Events from Kali in Elasticsearch	Done	Discover, 53 docs (agent logs)
auditd and system events from Kali	To verify	Need a query on the auditd and system datasets
Windows process events (Sysmon)	Done	IT-machine, event.code 1, 13 docs
Windows Security logons	Done	WIN-MKSP5LUC3TN, system.security, event.code 4624, 70 docs
FortiGate file reaches Elasticsearch	Done	Discover, 26 docs, log.file.path /var/log/fortigate.log
FortiGate fields parsed to ECS	To verify	Need fresh capture, or switch to the Fortinet integration
Disk growth measured	Done	3 MB in 43 s, about 5.9 GB per day (worst case)
Static IPs for server and agents	To do	Windows Server moved from .91 to .116
ILM policy applied	To do	soc-lab-2d, before detection rules go live
TLS with a trusted certificate	To do	Enrollment currently uses --insecure

10. Action Plan

This section records who did what in Week 1. The team is split by responsibility: documentation, Elastic Stack setup and lab infrastructure.

10.1 Team and responsibilities

Team member	Team	Week 1 responsibility
Mahmoud Ahmed Fathy Qamrahom Shalaby	Documentation	Wrote this report: architecture, data source inventory, log onboarding evidence, field mapping, retention plan and validation status.
Ahmed Mohamed Ahmed Mohamed Abdo	Elastic team	Set up Elastic (Fleet Server, Elasticsearch, Kibana), connected the agents to Elastic and set up the integrations.
Mohamed Emad Fathy Shabl	Elastic team	Set up Elastic (Fleet Server, Elasticsearch, Kibana), connected the agents to Elastic and set up the integrations.
Omnia Ali Mohamed Khallaf	Elastic team	Set up Elastic (Fleet Server, Elasticsearch, Kibana), connected the agents to Elastic and set up the integrations.
Mohamed Abdelhamid Abdelgalil Nagm	Infrastructure team	Set up the four lab VMs: FortiGate, Windows Server 2019, Windows 10 and Kali Linux.
Ahmed Ashraf Abdelhamid Mostafa Kandil	Infrastructure team	Set up the four lab VMs: FortiGate, Windows Server 2019, Windows 10 and Kali Linux.

10.2 Week 1 actions completed

Action	Owner	Result	Section
Set up the four VMs (FortiGate, Windows Server 2019, Windows 10, Kali)	Infrastructure team	Four VMs running on 192.168.1.0/24 and reachable from the SIEM server	3
Install and configure Fleet Server, Elasticsearch and Kibana on ubuntu-siem	Elastic team	Fleet Server on port 8220, Kibana on port 5601	3, 4
Enroll Elastic Agents on Windows 10, Windows Server and Kali	Elastic team	All four agents Healthy on version 8.19.23	6.1 to 6.4
Set up integrations and policies (system, Windows, Sysmon, auditd)	Elastic team	Sysmon, Windows Security and agent logs visible in Discover	5, 6
Send FortiGate logs to a file read by Filebeat	Elastic team and infrastructure team	26 documents from /var/log/fortigate.log in Discover	6.5
Write the Week 1 report and collect the evidence	Documentation	Report covering the five Week 1 deliverables	All